

# NanoGT Match Report: Kohlschutter-Tonz syndrome

**Disease:** Kohlschutter-Tonz syndrome (ORPHA:1946)

**Primary gene:** ROGDI

**Gene CDS:** 861 bp

**Inheritance:** Autosomal recessive

**Target tissues:** CNS

## Top 5 GT Precedent Matches

| Rank | Program    | Vector | Score  | Confidence      | Approval |
|------|------------|--------|--------|-----------------|----------|
| 1    | OAV101-IT  | AAV9   | 7.6/10 | [GREEN] High    | approved |
| 2    | Zolgensma  | AAV9   | 7.6/10 | [GREEN] High    | approved |
| 3    | BMN 307    | AAV5   | 7.3/10 | [YELLOW] Medium | phase2   |
| 4    | Libmeldy   | LV     | 7.3/10 | [YELLOW] Medium | approved |
| 5    | Strimvelis | LV     | 7.2/10 | [YELLOW] Medium | approved |

## Match #1: OAV101-IT

**Precedent disease:** Spinal Muscular Atrophy

**Vector:** AAV9

**Tissue target:** CNS/spinal cord

**Composite score:** 7.6 / 10

## Score Breakdown

| Dimension                 | Score | Max | What it measures                                     |
|---------------------------|-------|-----|------------------------------------------------------|
| Packaging fit             | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 2.00  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class             | 1.50  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 1.00  | 2.0 | Same or related biological pathway                   |
| Inheritance compatibility | 1.00  | 1.0 | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00  | 1.0 | Regulatory approval / trial stage                    |
| Immunogenicity            | 1.00  | 2.0 | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50  | 2.0 | Can GT be given before irreversible damage?          |

| Dimension                 | Score       | Max         | What it measures                                     |
|---------------------------|-------------|-------------|------------------------------------------------------|
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue            |
| Promoter availability     | 0.80        | 1.0         | Validated tissue-specific promoters exist            |
| Route of administration   | 0.70        | 1.0         | Established delivery route to target tissue          |
| <b>TOTAL (normalised)</b> | <b>7.56</b> | <b>10.0</b> | Raw sum / 18 × 10                                    |

## Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

## Match #2: Zolgensma

**Precedent disease:** Spinal Muscular Atrophy

**Vector:** AAV9

**Tissue target:** CNS/motor neuron

**Composite score:** 7.6 / 10

## Score Breakdown

| Dimension      | Score | Max | What it measures                               |
|----------------|-------|-----|------------------------------------------------|
| Packaging fit  | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity         |
| Tissue tropism | 2.00  | 2.0 | Vector naturally reaches disease target tissue |

| Dimension                 | Score       | Max         | What it measures                                      |
|---------------------------|-------------|-------------|-------------------------------------------------------|
| Protein class             | 1.50        | 2.0         | Same se-creted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 1.00        | 2.0         | Same or related biological pathway                    |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                  |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                     |
| Immunogenicity            | 1.00        | 2.0         | Pre-existing NAb seroprevalence for this vector       |
| Therapeutic window        | 1.50        | 2.0         | Can GT be given before irreversible damage?           |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours?  |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue             |
| Promoter availability     | 0.80        | 1.0         | Validated tissue-specific promoters exist             |
| Route of administration   | 0.70        | 1.0         | Established delivery route to target tissue           |
| <b>TOTAL (normalised)</b> | <b>7.56</b> | <b>10.0</b> | Raw sum / 18 × 10                                     |

## Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

### Match #3: BMN 307

**Precedent disease:** Phenylketonuria

**Vector:** AAV5

**Tissue target:** liver

**Composite score:** 7.3 / 10

#### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                     |
|---------------------------|-------------|-------------|------------------------------------------------------|
| Packaging fit             | 2.00        | 2.0         | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 1.00        | 2.0         | Vector naturally reaches disease target tissue       |
| Protein class             | 1.50        | 2.0         | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 1.00        | 2.0         | Same or related biological pathway                   |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 0.60        | 1.0         | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50        | 2.0         | Can GT be given before irreversible damage?          |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue            |
| Promoter availability     | 0.80        | 1.0         | Validated tissue-specific promoters exist            |
| Route of administration   | 0.70        | 1.0         | Established delivery route to target tissue          |
| <b>TOTAL (normalised)</b> | <b>7.33</b> | <b>10.0</b> | Raw sum / 18 × 10                                    |

#### Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

## Match #4: Libmeldy

**Precedent disease:** Metachromatic leukodystrophy

**Vector:** LV

**Tissue target:** hematopoietic/CNS

**Composite score:** 7.3 / 10

### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                     |
|---------------------------|-------------|-------------|------------------------------------------------------|
| Packaging fit             | 2.00        | 2.0         | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 1.50        | 2.0         | Vector naturally reaches disease target tissue       |
| Protein class             | 0.50        | 2.0         | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 1.00        | 2.0         | Same or related biological pathway                   |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50        | 2.0         | Can GT be given before irreversible damage?          |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue            |
| Promoter availability     | 0.80        | 1.0         | Validated tissue-specific promoters exist            |
| Route of administration   | 0.70        | 1.0         | Established delivery route to target tissue          |
| <b>TOTAL (normalised)</b> | <b>7.28</b> | <b>10.0</b> | Raw sum / $18 \times 10$                             |

## Rationale

- Gene CDS 861bp / cargo 8000bp (11% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

## Match #5: Strimvelis

**Precedent disease:** ADA-SCID

**Vector:** LV

**Tissue target:** hematopoietic

**Composite score:** 7.2 / 10

## Score Breakdown

| Dimension                 | Score | Max | What it measures                                     |
|---------------------------|-------|-----|------------------------------------------------------|
| Packaging fit             | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 0.30  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class             | 1.50  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 1.00  | 2.0 | Same or related biological pathway                   |
| Inheritance compatibility | 1.00  | 1.0 | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00  | 1.0 | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00  | 2.0 | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50  | 2.0 | Can GT be given before irreversible damage?          |

| Dimension                 | Score       | Max         | What it measures                                     |
|---------------------------|-------------|-------------|------------------------------------------------------|
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue            |
| Promoter availability     | 0.80        | 1.0         | Validated tissue-specific promoters exist            |
| Route of administration   | 0.70        | 1.0         | Established delivery route to target tissue          |
| <b>TOTAL (normalised)</b> | <b>7.17</b> | <b>10.0</b> | Raw sum / 18 × 10                                    |

## Rationale

- Gene CDS 861bp / cargo 8000bp (11% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk